

UNREAL EASING NODES

WHY EASE?

- Movement
- Cameras
- Input
- Scripted events
- Simple actions (door closing etc).
- Particle systems

INTERPOLATION NODES IN UNREAL

- When making our games, we often need to move or rotate an object from one position/rotation to another.
- We have several options for how to do this and each gives us a different gameplay effect and “feel”
- Understanding these concepts is the key to having your movement smooth and controlled. It also allows you to adjust the “Feeling” of movement.
- We will examine movement first, as it is simpler to understand, however the same principles apply to rotations.

EASING BASICS

- Basic, no easing
- For the sake of our situation, lets imagine a scenario where we have an object at position A and we want it to be at position B.
- The most basic way to do this would simply be to set the position of the object to the value B. This would effectively teleport the object without any intervening frames.



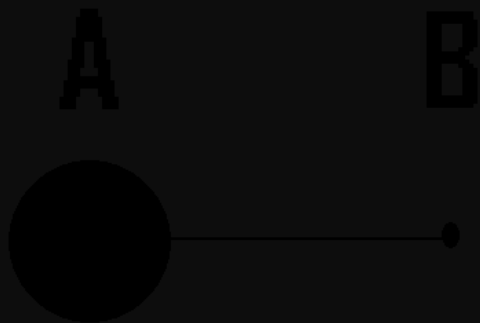
BACKGROUND: LINEAR INTERPOLATION

- To interpolate between two positions smoothly, having the same spacing between frames.
- This would be an object moving at a fixed velocity



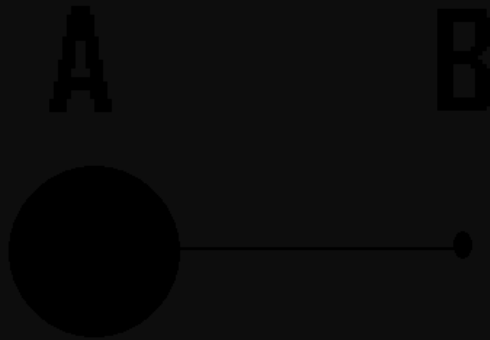
BACKGROUND: LINEAR INTERPOLATION

- We can also have an acceleration or deceleration applied, this is sometimes called “Easing In” and “Easing Out”.



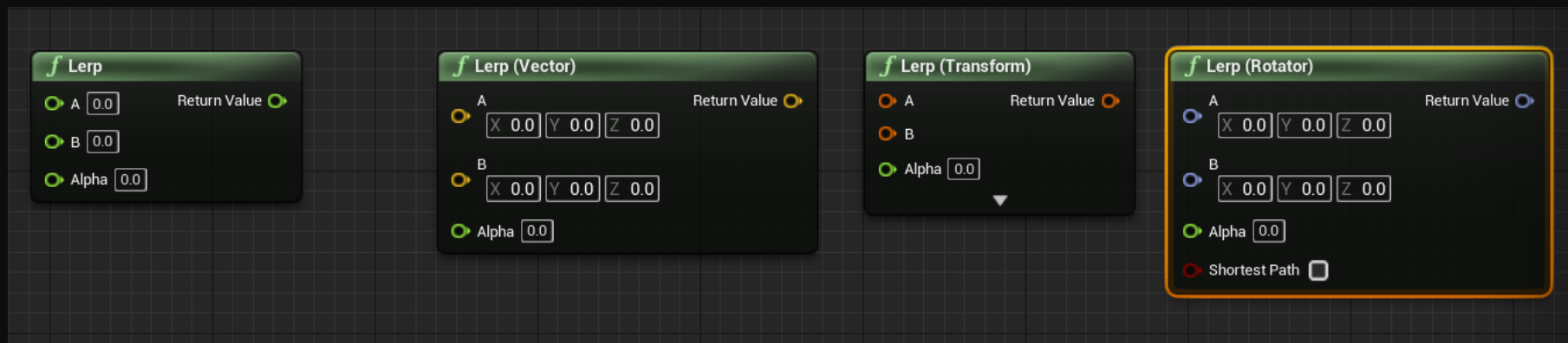
BACKGROUND: LINEAR INTERPOLATION

- If we accelerate and decelerate, we have an “EaseInOut” type movement.



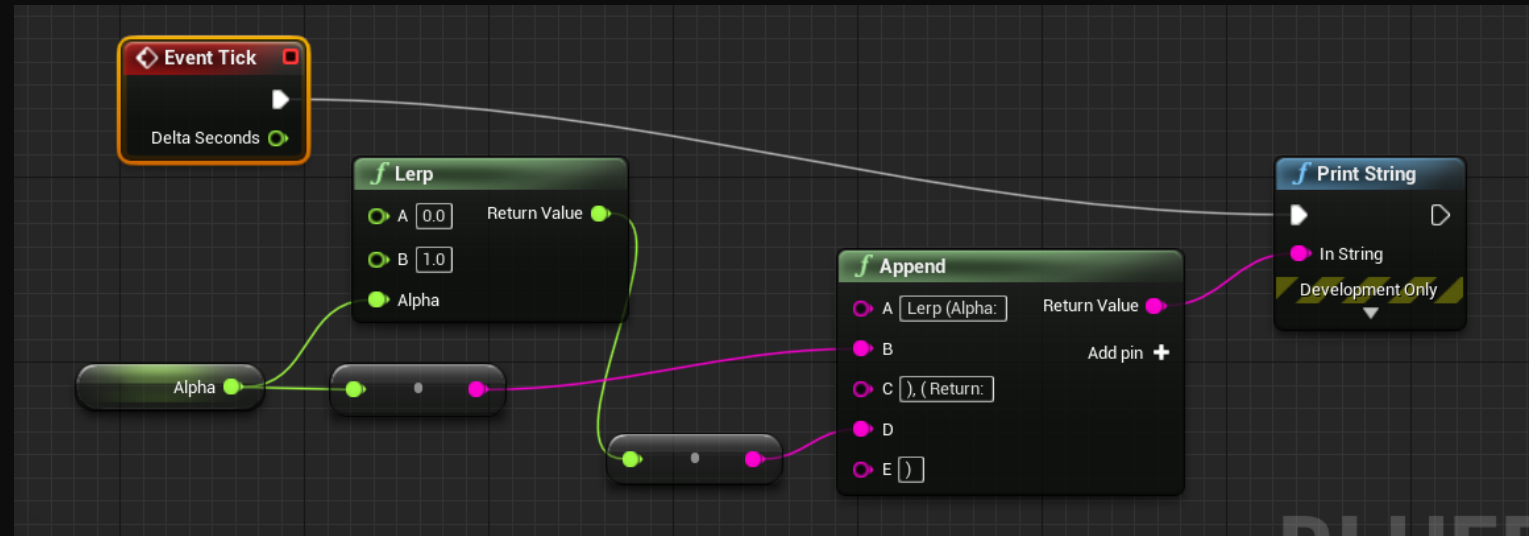
USEFUL UNREAL NODES: LERP

- The mysterious Lerp node can be tricky to get your head around at first glance, however it is very powerful and useful tool for controlling any number of different movements, rotations and transitions within unreal.
- A Lerp is short for “Linear Interpolation” and as we have seen, it is useful for transitioning between two states.
- There are several varieties of Lerp nodes, each dealing with a different data type, however they all work under the same principles.



USEFUL UNREAL NODES: LERP

- Basically, the lerp node accepts a starting value and an ending value (A & B) as well as the Alpha.
- The alpha specifies what proportion of the transition to display.
- Using a float for example, our simple example demonstrates what is really happening.

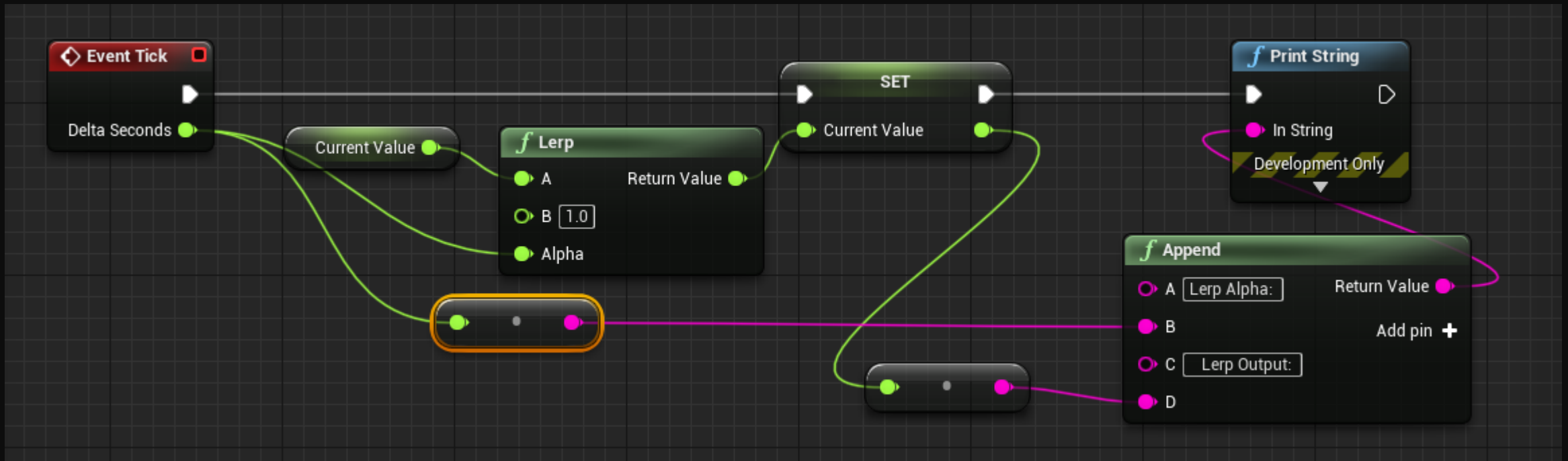


USEFUL UNREAL NODES: LERP

- Experiment with this example and what you will find is:
 - When the value of Alpha is 0, the output is 0
 - When the value of Alpha is 1, the output is 1
 - When the value of Alpha is 0.5, the output is 0.5
- Simply put, the Lerp is finding an interpolation of the range (A,B)
- An alpha of .5 means give us the value 50% between A and B
- An alpha of .75 means give us the value 75% between A and B, or .75 in this example.

USEFUL UNREAL NODES: LERP

- You might find this confusing as you often see a Lerp used with an input value of DeltaTime, something like this:
- To understand what this is really doing however, just think about what we talked about earlier.



USEFUL UNREAL NODES: LERP

- Here is some partial output for reference..
- The behaviour of this is that it will start updating the value quite quickly, and gradually slow to smaller and smaller output steps, approaching our B target vector but not *Really* reaching it.

[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.016667
[LerpBasics_2]	Lerp Alpha: 0.116648	Lerp Output: 0.131371
[LerpBasics_2]	Lerp Alpha: 0.063707	Lerp Output: 0.186709
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.200264
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.213593
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.2267
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.239589
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.252262
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.264725
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.27698
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.28903
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.30088
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.312532
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.32399
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.335257
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.346336
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.357231
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.367944
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.378479
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.388837
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.399024
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.40904
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.41889
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.428575
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.438099
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.447464
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.456673
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.465729
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.474633
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.48339
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.492
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.500467
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.508793
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.516979
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.52503
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.532946
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.54073
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.548385
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.555912
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.563314
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.570592

USEFUL UNREAL NODES: LERP

- This is more of the log as it approaches 1
- Note it doesn't actually reach the value of 1
 - (It will eventually but it will take a while!)
- This is similar to (although not exactly) the Logarithmic curve we examined earlier.

[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.978832
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.979185
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.979532
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.979873
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.980208
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.980538
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.980863
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.981182
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.981495
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.981804
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.982107
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.982405
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.982698
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.982987
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.98327
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.983549
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.983823
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.984093
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.984358
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.984619
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.984875
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.985127
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.985375
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.985619
[LerpBasics_2]	Lerp Alpha: 0.01667	Lerp Output: 0.985859
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.986094
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.986326
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.986554
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.986778
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.986998
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.987215
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.987428
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.987638
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.987844
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.988046
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.988246
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.988441
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.988634
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.988824
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.98901
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.989193
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.989373
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.98955
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.989724
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.989896
[LerpBasics_2]	Lerp Alpha: 0.016667	Lerp Output: 0.990064

USEFUL UNREAL NODES: LERP

- To Demonstrate what is happening here, I will use a simplified example with smaller time steps. Imagine a Lerp node with a value of $A = 0$, $B=1$. That is we are Lerping from 0 to 1. (We could use any values here but this is much simpler to understand). Imagine our Lerp Alpha (timestep) is 0.1
- At timestep 0 (t_0) our output will be 0, since the lerp has not been applied yet.
- At T_1 , our output will be 0.1, since we are .1 of the way from A to B.
- Since we are saving our output value back into our variable (that feeds into A), when we take T_2 , our A value will be 0.1
- This means that even though the Alpha stays the same, the range it is lerping over has changes from 0..1 to (0.1... 1).
- 0.1 of that range is 0.19, or 10% of the way from $A = 0.1$ to $B = 1.0$

USEFUL UNREAL NODES: LERP

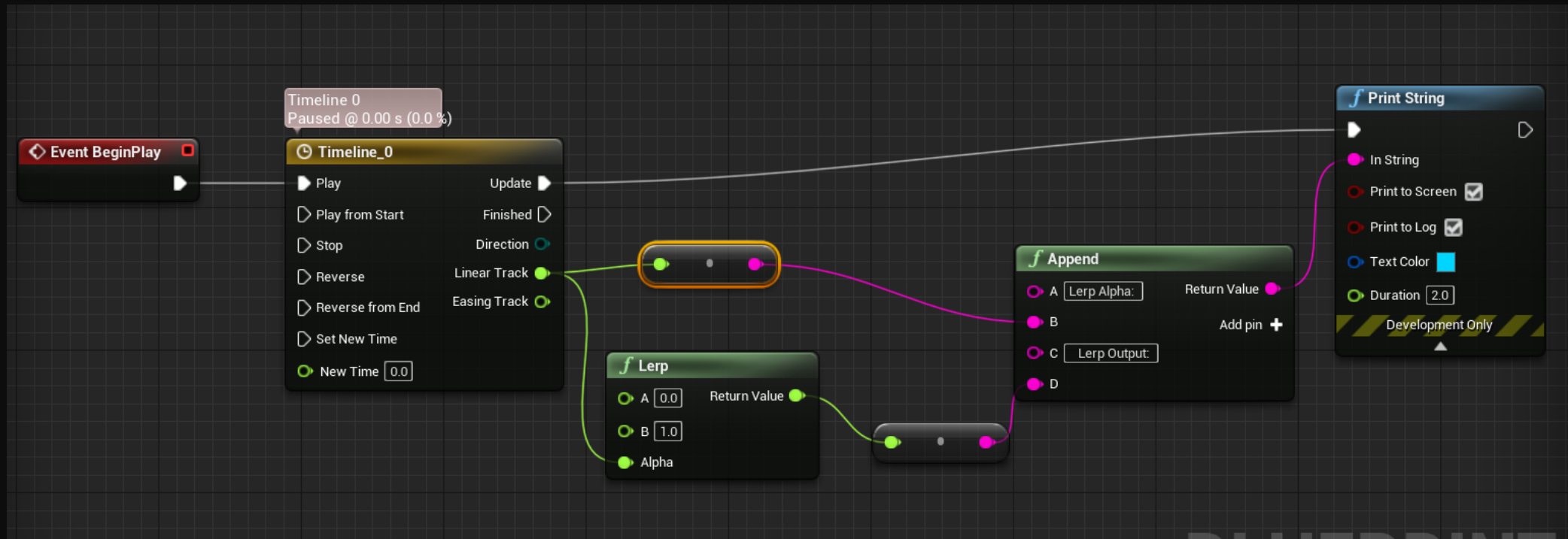
- If we continue running our experiment, we can see that we have a similar effect to what we had before.
- Question: if you don't save the output, what would you get?
 - The output would be 0.1 every frame since A would remain 0

Lerp Alpha: 0.1	Lerp Output: 0.1
Lerp Alpha: 0.1	Lerp Output: 0.19
Lerp Alpha: 0.1	Lerp Output: 0.271
Lerp Alpha: 0.1	Lerp Output: 0.3439
Lerp Alpha: 0.1	Lerp Output: 0.40951
Lerp Alpha: 0.1	Lerp Output: 0.468559
Lerp Alpha: 0.1	Lerp Output: 0.521703
Lerp Alpha: 0.1	Lerp Output: 0.569533
Lerp Alpha: 0.1	Lerp Output: 0.61258
Lerp Alpha: 0.1	Lerp Output: 0.651322
Lerp Alpha: 0.1	Lerp Output: 0.686189
Lerp Alpha: 0.1	Lerp Output: 0.71757
Lerp Alpha: 0.1	Lerp Output: 0.745813
Lerp Alpha: 0.1	Lerp Output: 0.771232
Lerp Alpha: 0.1	Lerp Output: 0.794109
Lerp Alpha: 0.1	Lerp Output: 0.814698
Lerp Alpha: 0.1	Lerp Output: 0.833228
Lerp Alpha: 0.1	Lerp Output: 0.849905
Lerp Alpha: 0.1	Lerp Output: 0.864915
Lerp Alpha: 0.1	Lerp Output: 0.878423
Lerp Alpha: 0.1	Lerp Output: 0.890581
Lerp Alpha: 0.1	Lerp Output: 0.901523
Lerp Alpha: 0.1	Lerp Output: 0.911371
Lerp Alpha: 0.1	Lerp Output: 0.920234
Lerp Alpha: 0.1	Lerp Output: 0.92821
Lerp Alpha: 0.1	Lerp Output: 0.935389
Lerp Alpha: 0.1	Lerp Output: 0.94185
Lerp Alpha: 0.1	Lerp Output: 0.947665
Lerp Alpha: 0.1	Lerp Output: 0.952899
Lerp Alpha: 0.1	Lerp Output: 0.957609
Lerp Alpha: 0.1	Lerp Output: 0.961848

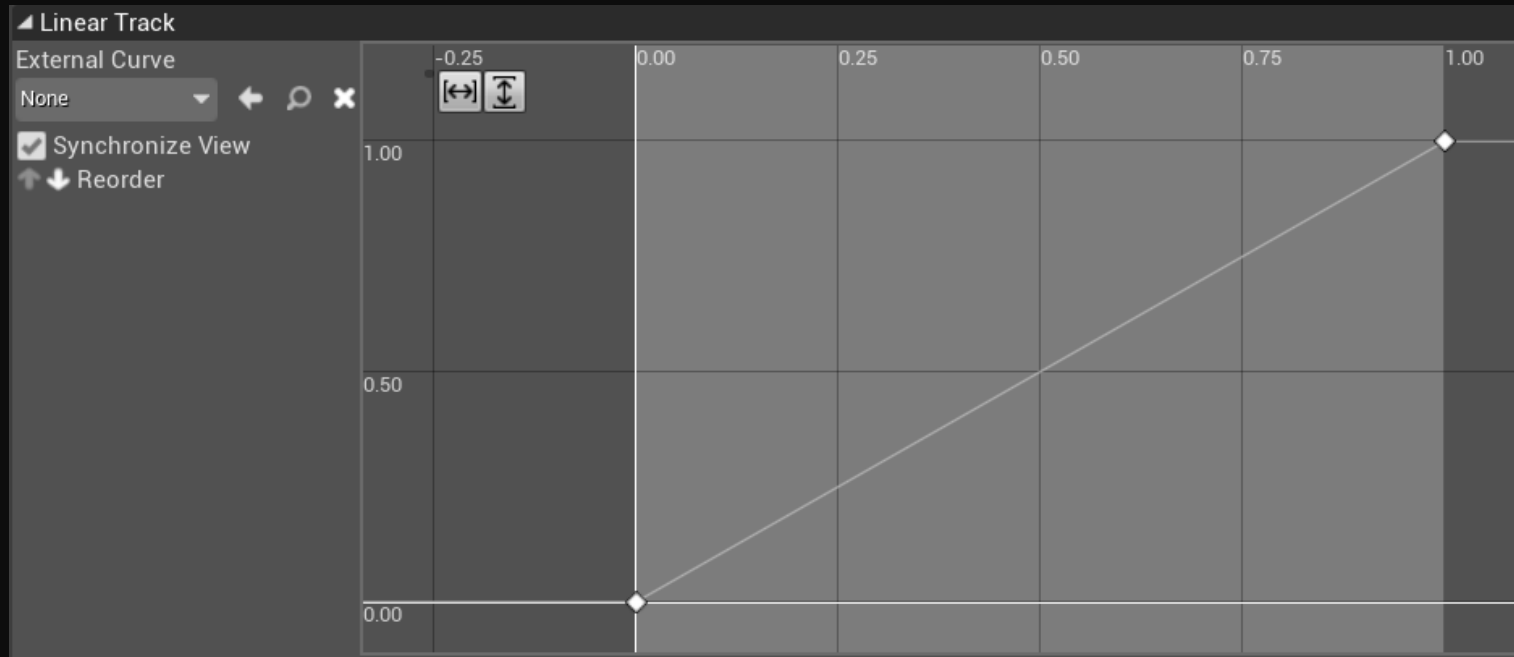
USEFUL UNREAL NODES: LERP

- In summary, Lerp often gets abused and misused. It is not a great option for using with deltaTime since it never quite reaches its target and there are better options for easing (as we will see).
- What lerp is extremely useful for is manually interpolating between two values, finding a midpoint or proportion of two values.
- While this can be used in many different ways, one of the most powerful ways of using this in Unreal is to combine the Lerp node with a Timeline. This way, you can control the easing precisely.
 - (Demo on next slide).

USEFUL UNREAL NODES: LERP WITH TIMELINE



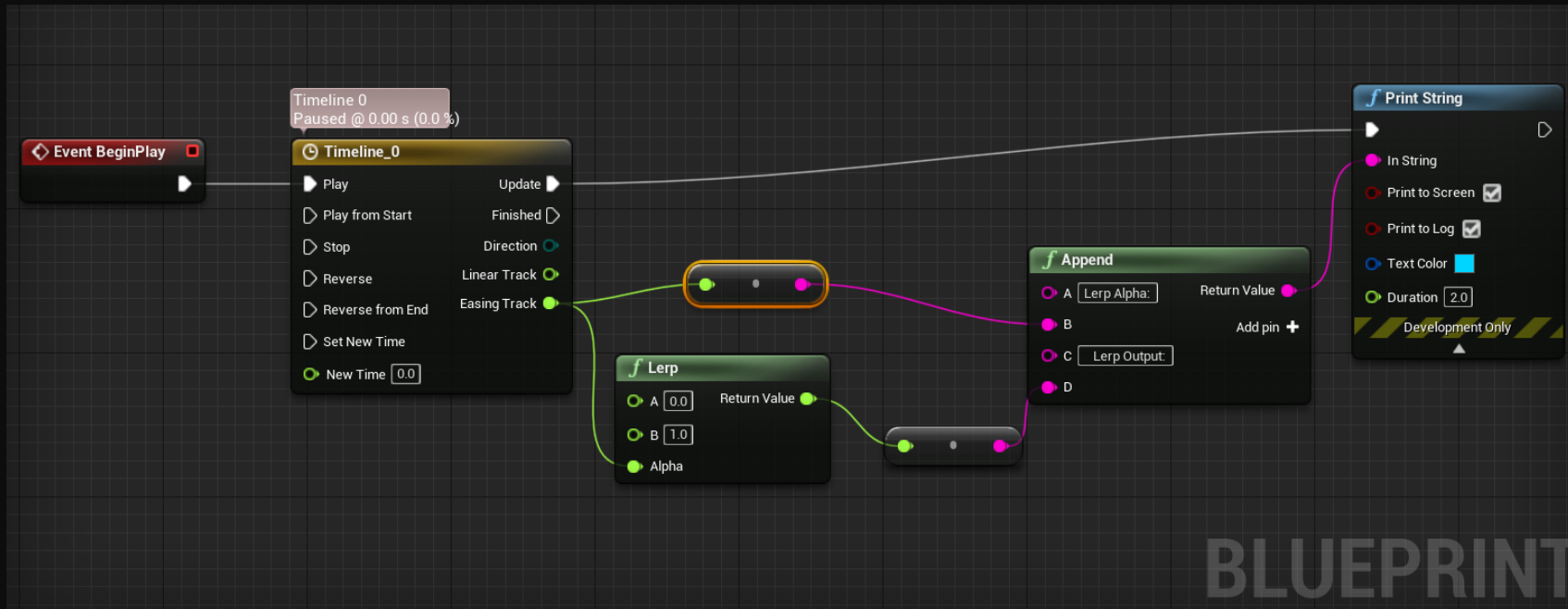
USEFUL UNREAL NODES: LERP WITH TIMELINE



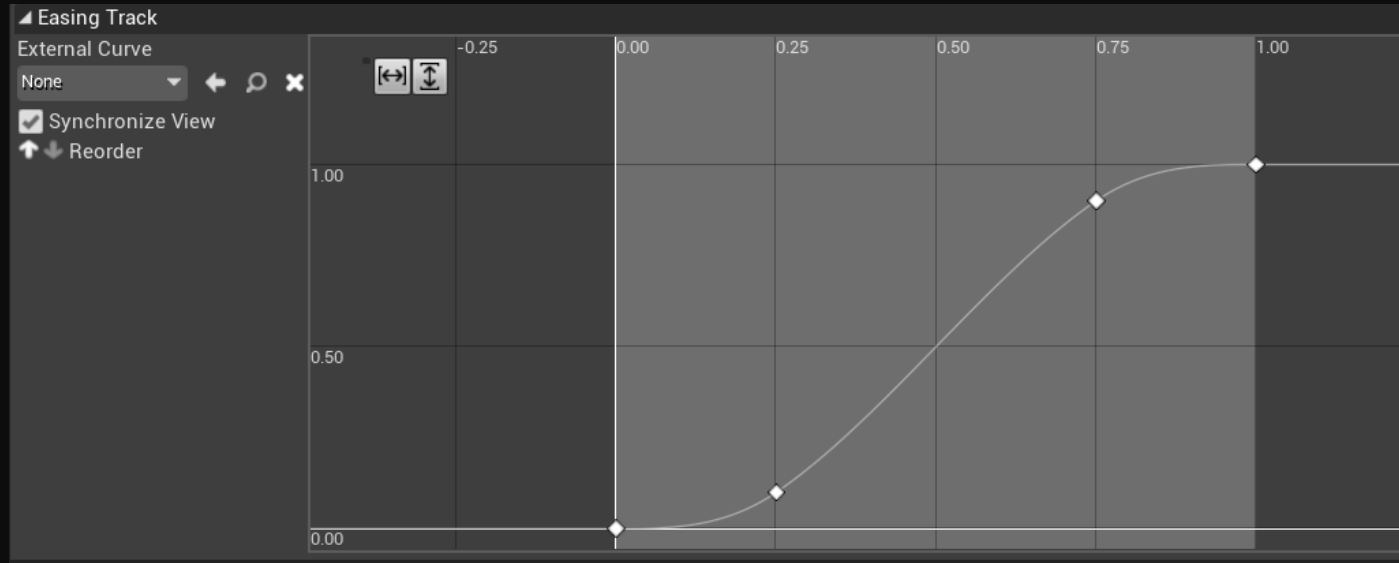
Lerp Alpha: 0.016667	Lerp Output: 0.016667
Lerp Alpha: 0.134483	Lerp Output: 0.134483
Lerp Alpha: 0.151115	Lerp Output: 0.151115
Lerp Alpha: 0.167817	Lerp Output: 0.167817
Lerp Alpha: 0.184483	Lerp Output: 0.184483
Lerp Alpha: 0.201115	Lerp Output: 0.201115
Lerp Alpha: 0.217817	Lerp Output: 0.217817
Lerp Alpha: 0.234484	Lerp Output: 0.234484
Lerp Alpha: 0.251151	Lerp Output: 0.251151
Lerp Alpha: 0.267818	Lerp Output: 0.267818
Lerp Alpha: 0.284485	Lerp Output: 0.284485
Lerp Alpha: 0.301152	Lerp Output: 0.301152
Lerp Alpha: 0.317819	Lerp Output: 0.317819
Lerp Alpha: 0.334485	Lerp Output: 0.334485
Lerp Alpha: 0.351152	Lerp Output: 0.351152
Lerp Alpha: 0.367819	Lerp Output: 0.367819
Lerp Alpha: 0.384486	Lerp Output: 0.384486
Lerp Alpha: 0.401153	Lerp Output: 0.401153
Lerp Alpha: 0.41782	Lerp Output: 0.41782
Lerp Alpha: 0.434487	Lerp Output: 0.434487
Lerp Alpha: 0.451154	Lerp Output: 0.451154
Lerp Alpha: 0.467821	Lerp Output: 0.467821
Lerp Alpha: 0.484488	Lerp Output: 0.484488
Lerp Alpha: 0.501155	Lerp Output: 0.501155
Lerp Alpha: 0.517822	Lerp Output: 0.517822
Lerp Alpha: 0.534488	Lerp Output: 0.534488
Lerp Alpha: 0.551155	Lerp Output: 0.551155
Lerp Alpha: 0.567822	Lerp Output: 0.567822
Lerp Alpha: 0.584489	Lerp Output: 0.584489

USEFUL UNREAL NODES: LERP WITH EASING TIMELINE

AS YOU CAN SEE, THIS GIVES US THE EASING WE WANT WHILE MAINTAINING SPECIFIC CONTROL OVER HOW IT FUNCTIONS. THIS EASING TRACK HAS A BIT OF AN S-CURVE SHAPE THAT WILL GIVE US AN EASE-IN AND EASE-OUT TYPE BEHAVIOUR.



USEFUL UNREAL NODES: LERP WITH EASING TIMELINE



Lerp Alpha: 0.00003	Lerp Output: 0.00003
Lerp Alpha: 0.085473	Lerp Output: 0.085473
Lerp Alpha: 0.122855	Lerp Output: 0.122855
Lerp Alpha: 0.144876	Lerp Output: 0.144876
Lerp Alpha: 0.168042	Lerp Output: 0.168042
Lerp Alpha: 0.192266	Lerp Output: 0.192266
Lerp Alpha: 0.217459	Lerp Output: 0.217459
Lerp Alpha: 0.243531	Lerp Output: 0.243531
Lerp Alpha: 0.270394	Lerp Output: 0.270394
Lerp Alpha: 0.29796	Lerp Output: 0.29796
Lerp Alpha: 0.326138	Lerp Output: 0.326138
Lerp Alpha: 0.35484	Lerp Output: 0.35484
Lerp Alpha: 0.384021	Lerp Output: 0.384021
Lerp Alpha: 0.413506	Lerp Output: 0.413506
Lerp Alpha: 0.443248	Lerp Output: 0.443248
Lerp Alpha: 0.473169	Lerp Output: 0.473169
Lerp Alpha: 0.503158	Lerp Output: 0.503158
Lerp Alpha: 0.533139	Lerp Output: 0.533139
Lerp Alpha: 0.563022	Lerp Output: 0.563022
Lerp Alpha: 0.592717	Lerp Output: 0.592717
Lerp Alpha: 0.622136	Lerp Output: 0.622136
Lerp Alpha: 0.651191	Lerp Output: 0.651191
Lerp Alpha: 0.679791	Lerp Output: 0.679791
Lerp Alpha: 0.707849	Lerp Output: 0.707849
Lerp Alpha: 0.735275	Lerp Output: 0.735275
Lerp Alpha: 0.761981	Lerp Output: 0.761981
Lerp Alpha: 0.787877	Lerp Output: 0.787877
Lerp Alpha: 0.812875	Lerp Output: 0.812875
Lerp Alpha: 0.836885	Lerp Output: 0.836885
Lerp Alpha: 0.85982	Lerp Output: 0.85982
Lerp Alpha: 0.881589	Lerp Output: 0.881589
Lerp Alpha: 0.902098	Lerp Output: 0.902098
Lerp Alpha: 0.935442	Lerp Output: 0.935442
Lerp Alpha: 0.949259	Lerp Output: 0.949259
Lerp Alpha: 0.960949	Lerp Output: 0.960949
Lerp Alpha: 0.970689	Lerp Output: 0.970689
Lerp Alpha: 0.978658	Lerp Output: 0.978658
Lerp Alpha: 0.985034	Lerp Output: 0.985034
Lerp Alpha: 0.989993	Lerp Output: 0.989993
Lerp Alpha: 0.993715	Lerp Output: 0.993715
Lerp Alpha: 0.996376	Lerp Output: 0.996376
Lerp Alpha: 0.998155	Lerp Output: 0.998155
Lerp Alpha: 0.999229	Lerp Output: 0.999229
Lerp Alpha: 0.999776	Lerp Output: 0.999776
Lerp Alpha: 0.999973	Lerp Output: 0.999973
Lerp Alpha: 1.0	Lerp Output: 1.0

USEFUL UNREAL NODES: LERP

- As you can see, Lerp is extremely useful and can be used to interpolate between any number of different things, including Vectors, rotations and entire transforms.
- It is also one of the most likely to be misunderstood and misused so it is important to have a good idea of what it is doing
- It's also quite simple to understand once you get the basic concept
- Next we will look at some alternative nodes in Unreal for easing between different states.

USEFUL UNREAL NODES: LERP

- While Lerp will move you gradually closer to a point, there are some issues using it for this effect.
 - 1. It can be hard to control the rate at which it reaches its goal
 - 2. It doesn't quite reach the goal value, when tends to leave it sort of jittering as it inches towards it.
- As we will soon see, there are alternative nodes that achieve a similar effect but are better suited to this purpose.

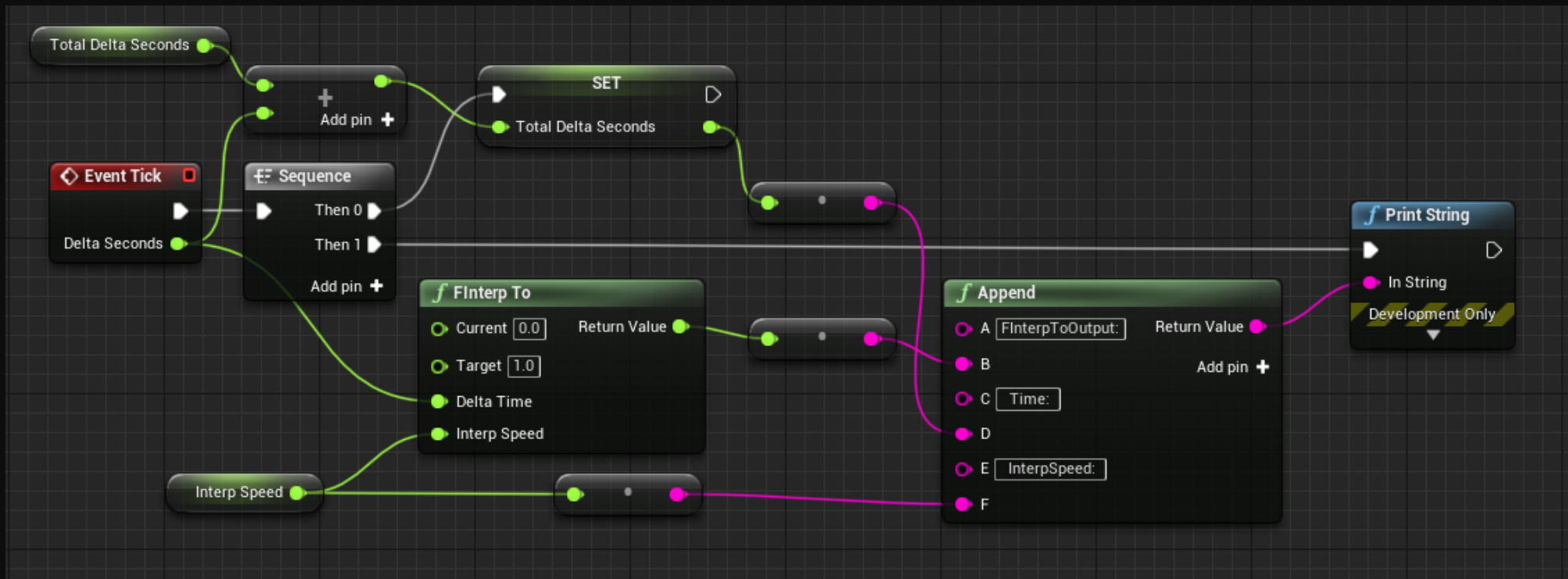
USEFUL UNREAL NODES: INTERPTO

- InterpTo is similar to Lerp, however it includes an input variable (InterpSpeed) specifically for adjusting the rate of change.
- It took some experimentation to figure out exactly what it is doing.



USEFUL UNREAL NODES: INTERPTO

- First Try: My initial attempt looked something like this:



USEFUL UNREAL NODES: INTERPTO

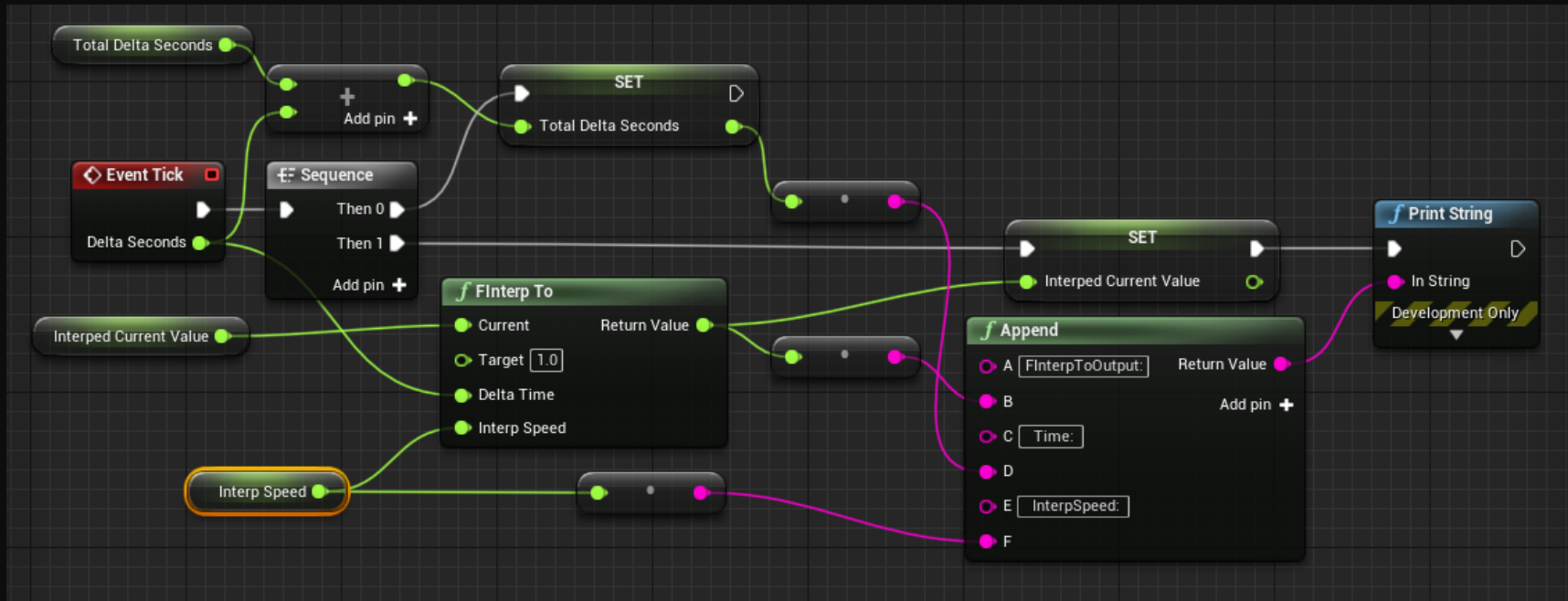
- First Try: However it gave me output like this:
- As you can see, a speed of 0 simply gave me the target value immediately, however the speed of 1 just gave me the same value over and over, not exactly useful.
- However, this reminded me of using a Lerp without passing the current value back in every frame, so making that adjustment gave me something more useful.
- Also, note but ignore the fluctuation in the first two lines, likely caused by the previous frame deltaTime being exceptionally small (due to it being the first frame in the scene I was running).

```
FInterpToOutput:1.0 Time: 0.016667 InterpSpeed: 0.0
FInterpToOutput:1.0 Time: 0.178357 InterpSpeed: 0.0
FInterpToOutput:1.0 Time: 0.195024 InterpSpeed: 0.0
FInterpToOutput:1.0 Time: 0.211691 InterpSpeed: 0.0
FInterpToOutput:1.0 Time: 0.228358 InterpSpeed: 0.0
FInterpToOutput:1.0 Time: 0.245025 InterpSpeed: 0.0
FInterpToOutput:1.0 Time: 0.261691 InterpSpeed: 0.0
FInterpToOutput:1.0 Time: 0.283521 InterpSpeed: 0.0
FInterpToOutput:1.0 Time: 0.300188 InterpSpeed: 0.0
FInterpToOutput:1.0 Time: 0.316855 InterpSpeed: 0.0
FInterpToOutput:1.0 Time: 0.333522 InterpSpeed: 0.0
FInterpToOutput:1.0 Time: 0.350189 InterpSpeed: 0.0
FInterpToOutput:1.0 Time: 0.366856 InterpSpeed: 0.0
FInterpToOutput:1.0 Time: 0.383523 InterpSpeed: 0.0
```

```
FInterpToOutput:0.016667 Time: 0.016667 InterpSpeed: 1.0
FInterpToOutput:0.099715 Time: 0.116381 InterpSpeed: 1.0
FInterpToOutput:0.027022 Time: 0.143403 InterpSpeed: 1.0
FInterpToOutput:0.016667 Time: 0.16007 InterpSpeed: 1.0
FInterpToOutput:0.016667 Time: 0.176737 InterpSpeed: 1.0
FInterpToOutput:0.016667 Time: 0.193404 InterpSpeed: 1.0
FInterpToOutput:0.016667 Time: 0.210071 InterpSpeed: 1.0
FInterpToOutput:0.016667 Time: 0.226738 InterpSpeed: 1.0
FInterpToOutput:0.016667 Time: 0.243405 InterpSpeed: 1.0
FInterpToOutput:0.016667 Time: 0.260072 InterpSpeed: 1.0
FInterpToOutput:0.016667 Time: 0.276739 InterpSpeed: 1.0
FInterpToOutput:0.016667 Time: 0.293406 InterpSpeed: 1.0
FInterpToOutput:0.016667 Time: 0.310073 InterpSpeed: 1.0
FInterpToOutput:0.016667 Time: 0.32674 InterpSpeed: 1.0
```

USEFUL UNREAL NODES: INTERPTO

- INCLUDING THE INPUT (INTERPEDCURRENTVALUE) FIXED THAT ISSUE AND GAVE ME THE OUTPUT I WAS EXPECTING



USEFUL UNREAL NODES: INTERPTO

- TRYING OUT SOME DIFFERENT SPEEDS GAVE INTERESTING RESULTS:

```
Time: 0.016667 InterpSpeed: 0.0
FInterpToOutput:1.0 Time: 0.177438 InterpSpeed: 0.0
FInterpToOutput:1.0 Time: 0.201968 InterpSpeed: 0.0
FInterpToOutput:1.0 Time: 0.218634 InterpSpeed: 0.0
FInterpToOutput:1.0 Time: 0.235302 InterpSpeed: 0.0
FInterpToOutput:1.0 Time: 0.251969 InterpSpeed: 0.0
FInterpToOutput:1.0 Time: 0.268636 InterpSpeed: 0.0
FInterpToOutput:1.0 Time: 0.285302 InterpSpeed: 0.0
FInterpToOutput:1.0 Time: 0.301969 InterpSpeed: 0.0
FInterpToOutput:1.0 Time: 0.318636 InterpSpeed: 0.0
FInterpToOutput:1.0 Time: 0.335303 InterpSpeed: 0.0
FInterpToOutput:1.0 Time: 0.35197 InterpSpeed: 0.0
FInterpToOutput:1.0 Time: 0.368638 InterpSpeed: 0.0
FInterpToOutput:1.0 Time: 0.385304 InterpSpeed: 0.0
FInterpToOutput:1.0 Time: 0.401971 InterpSpeed: 0.0
FInterpToOutput:1.0 Time: 0.418638 InterpSpeed: 0.0
FInterpToOutput:1.0 Time: 0.435305 InterpSpeed: 0.0
```

```
FInterpToOutput:0.033056 Time: 0.016667 InterpSpeed: 1.0
FInterpToOutput:0.215171 Time: 0.123285 InterpSpeed: 1.0
FInterpToOutput:0.162195 Time: 0.146717 InterpSpeed: 1.0
FInterpToOutput:0.170452 Time: 0.163384 InterpSpeed: 1.0
FInterpToOutput:0.184715 Time: 0.180314 InterpSpeed: 1.0
FInterpToOutput:0.198089 Time: 0.196982 InterpSpeed: 1.0
FInterpToOutput:0.211454 Time: 0.213648 InterpSpeed: 1.0
FInterpToOutput:0.224596 Time: 0.230315 InterpSpeed: 1.0
FInterpToOutput:0.23752 Time: 0.246982 InterpSpeed: 1.0
FInterpToOutput:0.250228 Time: 0.263649 InterpSpeed: 1.0
FInterpToOutput:0.262724 Time: 0.280316 InterpSpeed: 1.0
FInterpToOutput:0.275013 Time: 0.296983 InterpSpeed: 1.0
FInterpToOutput:0.287096 Time: 0.31365 InterpSpeed: 1.0
FInterpToOutput:0.298978 Time: 0.330316 InterpSpeed: 1.0
FInterpToOutput:0.310662 Time: 0.346983 InterpSpeed: 1.0
FInterpToOutput:0.322151 Time: 0.36365 InterpSpeed: 1.0
FInterpToOutput:0.333449 Time: 0.380317 InterpSpeed: 1.0
FInterpToOutput:0.344558 Time: 0.396985 InterpSpeed: 1.0
FInterpToOutput:0.355482 Time: 0.413651 InterpSpeed: 1.0
FInterpToOutput:0.366224 Time: 0.430318 InterpSpeed: 1.0
FInterpToOutput:0.376787 Time: 0.446985 InterpSpeed: 1.0
FInterpToOutput:0.387174 Time: 0.463652 InterpSpeed: 1.0
```


USEFUL UNREAL NODES: INTERPTO

- TRYING OUT SOME DIFFERENT SPEEDS GAVE INTERESTING RESULTS:

```
FInterpToOutput:0.065556 Time: 0.016667 InterpSpeed: 2.0
FInterpToOutput:0.574537 Time: 0.184954 InterpSpeed: 2.0
FInterpToOutput:0.435166 Time: 0.215713 InterpSpeed: 2.0
FInterpToOutput:0.437597 Time: 0.23238 InterpSpeed: 2.0
FInterpToOutput:0.456344 Time: 0.249047 InterpSpeed: 2.0
FInterpToOutput:0.474586 Time: 0.265769 InterpSpeed: 2.0
FInterpToOutput:0.492042 Time: 0.282436 InterpSpeed: 2.0
FInterpToOutput:0.508975 Time: 0.299103 InterpSpeed: 2.0
FInterpToOutput:0.525372 Time: 0.315785 InterpSpeed: 2.0
FInterpToOutput:0.541179 Time: 0.332452 InterpSpeed: 2.0
FInterpToOutput:0.556473 Time: 0.349119 InterpSpeed: 2.0
FInterpToOutput:0.571257 Time: 0.365786 InterpSpeed: 2.0
FInterpToOutput:0.585548 Time: 0.382452 InterpSpeed: 2.0
FInterpToOutput:0.599364 Time: 0.399119 InterpSpeed: 2.0
FInterpToOutput:0.612718 Time: 0.415786 InterpSpeed: 2.0
FInterpToOutput:0.625628 Time: 0.432453 InterpSpeed: 2.0
FInterpToOutput:0.638107 Time: 0.44912 InterpSpeed: 2.0
FInterpToOutput:0.650171 Time: 0.465787 InterpSpeed: 2.0
FInterpToOutput:0.661832 Time: 0.482454 InterpSpeed: 2.0
FInterpToOutput:0.673104 Time: 0.499121 InterpSpeed: 2.0
```

```
FInterpToOutput:0.305559 Time: 0.016667 InterpSpeed: 10.0
FInterpToOutput:1.0 Time: 0.177296 InterpSpeed: 10.0
FInterpToOutput:1.0 Time: 0.193963 InterpSpeed: 10.0
FInterpToOutput:1.0 Time: 0.21063 InterpSpeed: 10.0
FInterpToOutput:1.0 Time: 0.227297 InterpSpeed: 10.0
FInterpToOutput:1.0 Time: 0.243964 InterpSpeed: 10.0
FInterpToOutput:1.0 Time: 0.260631 InterpSpeed: 10.0
FInterpToOutput:1.0 Time: 0.277298 InterpSpeed: 10.0
FInterpToOutput:1.0 Time: 0.293965 InterpSpeed: 10.0
FInterpToOutput:1.0 Time: 0.310632 InterpSpeed: 10.0
FInterpToOutput:1.0 Time: 0.327299 InterpSpeed: 10.0
FInterpToOutput:1.0 Time: 0.343966 InterpSpeed: 10.0
FInterpToOutput:1.0 Time: 0.360633 InterpSpeed: 10.0
FInterpToOutput:1.0 Time: 0.3773 InterpSpeed: 10.0
FInterpToOutput:1.0 Time: 0.393967 InterpSpeed: 10.0
FInterpToOutput:1.0 Time: 0.410634 InterpSpeed: 10.0
FInterpToOutput:1.0 Time: 0.427301 InterpSpeed: 10.0
FInterpToOutput:1.0 Time: 0.443968 InterpSpeed: 10.0
```


USEFUL UNREAL NODES: INTERPTO

- AS YOU CAN SEE, A SPEED OF 0 IMMEDIATELY SET THE OUTPUT TO THE TARGET VALUE
- A SPEED OF 10 WAS TOO FAST AND SET THE VALUE IMMEDIATELY TO THE TARGET VALUE AS WELL.
- IT DOESN'T SHOW UP WELL IN MY DATA SAMPLES, BUT A NICE FEATURE OF INTERPTO IS THAT IT DOES EVENTUALLY REACH THE TARGET VALUE.

INTERPTO QUESTIONS:

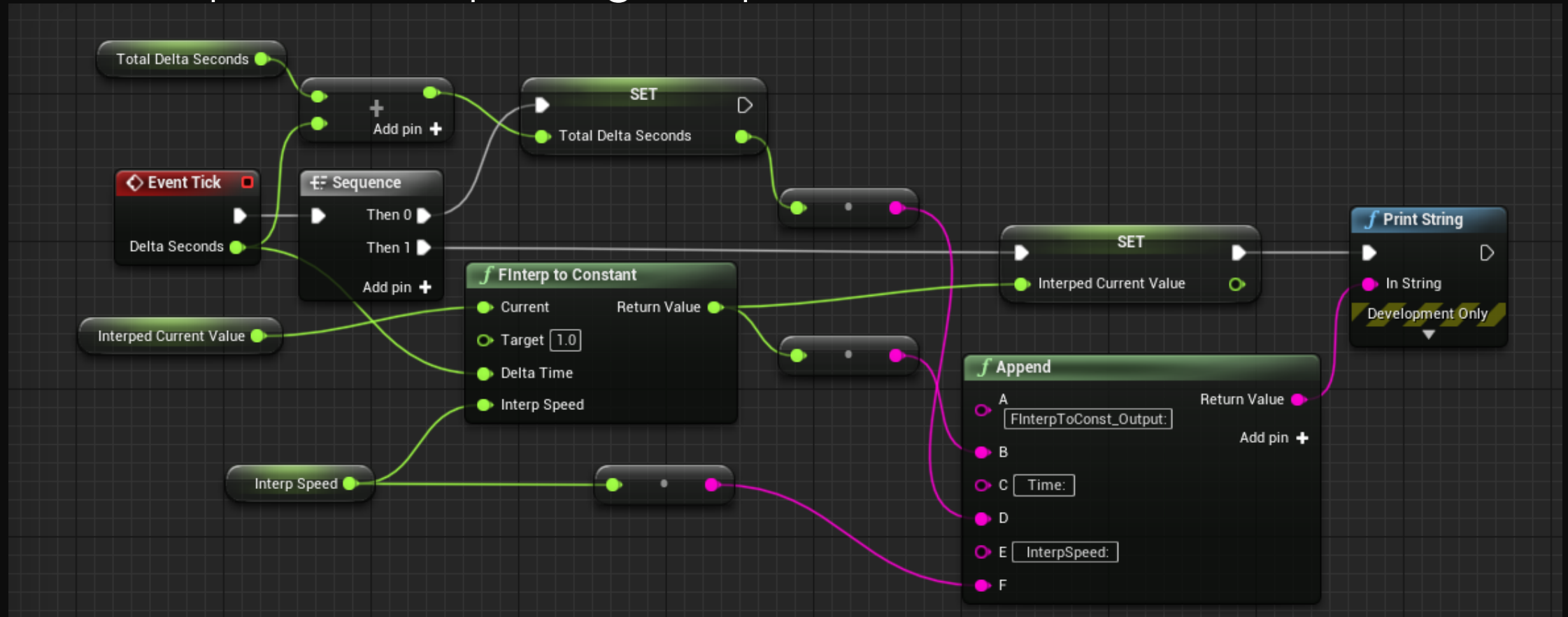
- IF YOU WERE TO GRAPH THE OUTPUT (Y AXIS) OVER THE TIME (X AXIS), WHICH TYPE OF EQUATION WOULD THIS FUNCTION RESEMBLE?
- EXPERIMENTING WITH THE INTERPTO NODE IN A SIMILAR SETUP, WHAT INTERPEDCURRENTVALUE WOULD GIVE YOU AN OUTPUT EQUAL TO THE TARGET VALUE OVER ~1 SECOND? ~5 SECONDS?
- DO SOME EXPERIMENTS TO FIND THE RELATIONSHIP BETWEEN INTERPEDCURRENTVALUE AND TRANSITION TIME? TRY SOME DIFFERENT VALUE AND RECORD HOW LONG IT TAKES TO REACH THE GOAL. IF YOU GRAPH THESE VALUES, DOES IT GIVE YOU A CONSISTENT CHANGE? WHAT TYPE OF GRAPH DOES IT GENERATE?

USEFUL UNREAL NODES: INTERPTOCONST

- Not really like InterpTo, despite the name.
- InterpToConst will try to reach a target value at a fixed rate.
- This means each step should be (close to) the same distance.
- Be careful as there are several differences between how InterpTo and InterpToConst work.

USEFUL UNREAL NODES: INTERPTOCONST

- I will use the exact same testing setup as before, with InterpToConst replacing InterpTo



USEFUL UNREAL NODES: INTERPTOCONST

- Some sample output shows some differences. What do you notice that is different from InterpTo?

```
FInterpToConst_Output:0.0 Time: 0.016667 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.165271 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.181938 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.198605 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.215272 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.232334 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.249001 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.265669 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.282336 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.299003 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.31567 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.332337 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.349004 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.365671 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.382338 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.399005 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.415671 InterpSpeed: 0.0
```

```
FInterpToConst_Output:0.033334 Time: 0.016667 InterpSpeed: 1.0
FInterpToConst_Output:0.313386 Time: 0.165026 InterpSpeed: 1.0
FInterpToConst_Output:0.218568 Time: 0.191797 InterpSpeed: 1.0
FInterpToConst_Output:0.234363 Time: 0.21308 InterpSpeed: 1.0
FInterpToConst_Output:0.246414 Time: 0.229747 InterpSpeed: 1.0
FInterpToConst_Output:0.263081 Time: 0.246414 InterpSpeed: 1.0
FInterpToConst_Output:0.279748 Time: 0.263081 InterpSpeed: 1.0
FInterpToConst_Output:0.296415 Time: 0.279748 InterpSpeed: 1.0
FInterpToConst_Output:0.313082 Time: 0.296415 InterpSpeed: 1.0
FInterpToConst_Output:0.329749 Time: 0.313082 InterpSpeed: 1.0
FInterpToConst_Output:0.346416 Time: 0.329749 InterpSpeed: 1.0
FInterpToConst_Output:0.363083 Time: 0.346416 InterpSpeed: 1.0
FInterpToConst_Output:0.37975 Time: 0.363083 InterpSpeed: 1.0
FInterpToConst_Output:0.396416 Time: 0.37975 InterpSpeed: 1.0
FInterpToConst_Output:0.413084 Time: 0.396417 InterpSpeed: 1.0
FInterpToConst_Output:0.42975 Time: 0.413083 InterpSpeed: 1.0
FInterpToConst_Output:0.446418 Time: 0.42975 InterpSpeed: 1.0
FInterpToConst_Output:0.463085 Time: 0.446418 InterpSpeed: 1.0
FInterpToConst_Output:0.479751 Time: 0.463084 InterpSpeed: 1.0
FInterpToConst_Output:0.496418 Time: 0.479751 InterpSpeed: 1.0
FInterpToConst_Output:0.513085 Time: 0.496418 InterpSpeed: 1.0
```

USEFUL UNREAL NODES: INTERPTOCONST

- Some sample output shows some differences.
What do you notice that is different from InterpTo? What is similar?

```
FInterpToConst_Output:0.066667 Time: 0.016667 InterpSpeed: 2.0
FInterpToConst_Output:0.652074 Time: 0.171352 InterpSpeed: 2.0
FInterpToConst_Output:0.453181 Time: 0.198971 InterpSpeed: 2.0
FInterpToConst_Output:0.464609 Time: 0.215638 InterpSpeed: 2.0
FInterpToConst_Output:0.497944 Time: 0.232305 InterpSpeed: 2.0
FInterpToConst_Output:0.531279 Time: 0.248972 InterpSpeed: 2.0
FInterpToConst_Output:0.564612 Time: 0.265639 InterpSpeed: 2.0
FInterpToConst_Output:0.597945 Time: 0.282306 InterpSpeed: 2.0
FInterpToConst_Output:0.631279 Time: 0.298973 InterpSpeed: 2.0
FInterpToConst_Output:0.664612 Time: 0.315639 InterpSpeed: 2.0
FInterpToConst_Output:0.697947 Time: 0.332307 InterpSpeed: 2.0
FInterpToConst_Output:0.731282 Time: 0.348974 InterpSpeed: 2.0
FInterpToConst_Output:0.764616 Time: 0.365641 InterpSpeed: 2.0
FInterpToConst_Output:0.79795 Time: 0.382308 InterpSpeed: 2.0
FInterpToConst_Output:0.831284 Time: 0.398975 InterpSpeed: 2.0
FInterpToConst_Output:0.864618 Time: 0.415642 InterpSpeed: 2.0
FInterpToConst_Output:0.897952 Time: 0.432309 InterpSpeed: 2.0
FInterpToConst_Output:0.931285 Time: 0.448976 InterpSpeed: 2.0
FInterpToConst_Output:0.964619 Time: 0.465643 InterpSpeed: 2.0
FInterpToConst_Output:0.997953 Time: 0.482309 InterpSpeed: 2.0
FInterpToConst_Output:1.0 Time: 0.498977 InterpSpeed: 2.0
FInterpToConst_Output:1.0 Time: 0.515644 InterpSpeed: 2.0
FInterpToConst_Output:1.0 Time: 0.532311 InterpSpeed: 2.0
FInterpToConst_Output:1.0 Time: 0.548978 InterpSpeed: 2.0
```

```
FInterpToConst_Output:0.923586 Time: 0.162265 InterpSpeed: 3.0
FInterpToConst_Output:0.686387 Time: 0.19553 InterpSpeed: 3.0
FInterpToConst_Output:0.686593 Time: 0.212197 InterpSpeed: 3.0
FInterpToConst_Output:0.736592 Time: 0.228864 InterpSpeed: 3.0
FInterpToConst_Output:0.786593 Time: 0.245531 InterpSpeed: 3.0
FInterpToConst_Output:0.836594 Time: 0.262198 InterpSpeed: 3.0
FInterpToConst_Output:0.886594 Time: 0.278865 InterpSpeed: 3.0
FInterpToConst_Output:0.936595 Time: 0.295531 InterpSpeed: 3.0
FInterpToConst_Output:0.986596 Time: 0.312198 InterpSpeed: 3.0
FInterpToConst_Output:1.0 Time: 0.328865 InterpSpeed: 3.0
FInterpToConst_Output:1.0 Time: 0.345532 InterpSpeed: 3.0
FInterpToConst_Output:1.0 Time: 0.362199 InterpSpeed: 3.0
FInterpToConst_Output:1.0 Time: 0.378866 InterpSpeed: 3.0
FInterpToConst_Output:1.0 Time: 0.395533 InterpSpeed: 3.0
FInterpToConst_Output:1.0 Time: 0.4122 InterpSpeed: 3.0
FInterpToConst_Output:1.0 Time: 0.428867 InterpSpeed: 3.0
```

```
FInterpToConst_Output:0.133336 Time: 0.016667 InterpSpeed: 4.0
FInterpToConst_Output:1.0 Time: 0.175352 InterpSpeed: 4.0
FInterpToConst_Output:1.0 Time: 0.215065 InterpSpeed: 4.0
FInterpToConst_Output:1.0 Time: 0.233001 InterpSpeed: 4.0
FInterpToConst_Output:1.0 Time: 0.249668 InterpSpeed: 4.0
FInterpToConst_Output:1.0 Time: 0.266335 InterpSpeed: 4.0
FInterpToConst_Output:1.0 Time: 0.283002 InterpSpeed: 4.0
FInterpToConst_Output:1.0 Time: 0.299668 InterpSpeed: 4.0
FInterpToConst_Output:1.0 Time: 0.316335 InterpSpeed: 4.0
```

USEFUL UNREAL NODES: INTERPTOCONST

- The first thing I noticed was that an InterpSpeed of 0 would not update the output at all.
- It is always useful to know how your nodes will deal with boundary conditions (such as a speed of 0). It is useful to note the difference in behaviour here.

```
FInterpToConst_Output:0.0 Time: 0.016667 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.165271 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.181938 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.198605 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.215272 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.232334 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.249001 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.265669 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.282336 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.299003 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.31567 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.332337 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.349004 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.365671 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.382338 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.399005 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.415671 InterpSpeed: 0.0
```


USEFUL UNREAL NODES: INTERPTOCONST

- The Second thing I noticed is the deltaTime in the first frame of running the program also threw off the consistency of the output value. If you skip the first value, all the rest of them are more or less equal distances apart (constant rate of change). This is also a bit of a glitch but useful to be aware of.

```
FInterpToConst_Output:0.0 Time: 0.016667 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.165271 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.181938 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.198605 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.215272 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.232334 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.249001 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.265669 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.282336 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.299003 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.31567 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.332337 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.349004 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.365671 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.382338 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.399005 InterpSpeed: 0.0
FInterpToConst_Output:0.0 Time: 0.415671 InterpSpeed: 0.0
```


USEFUL UNREAL NODES: INTERPTOCONST

- Comparing similar tests of the two nodes side by side show that they don't seem to have any relation in terms of speed. The InterpSpeed variables have totally different meanings, so there is no simple way to map from one to the other.

FInterpToOutput:0.033056	Time: 0.016667	InterpSpeed: 1.0
FInterpToOutput:0.215171	Time: 0.123285	InterpSpeed: 1.0
FInterpToOutput:0.162195	Time: 0.146717	InterpSpeed: 1.0
FInterpToOutput:0.170452	Time: 0.163384	InterpSpeed: 1.0
FInterpToOutput:0.184715	Time: 0.180314	InterpSpeed: 1.0
FInterpToOutput:0.198089	Time: 0.196982	InterpSpeed: 1.0
FInterpToOutput:0.211454	Time: 0.213648	InterpSpeed: 1.0
FInterpToOutput:0.224596	Time: 0.230315	InterpSpeed: 1.0
FInterpToOutput:0.23752	Time: 0.246982	InterpSpeed: 1.0
FInterpToOutput:0.250228	Time: 0.263649	InterpSpeed: 1.0
FInterpToOutput:0.262724	Time: 0.280316	InterpSpeed: 1.0
FInterpToOutput:0.275013	Time: 0.296983	InterpSpeed: 1.0
FInterpToOutput:0.287096	Time: 0.31365	InterpSpeed: 1.0
FInterpToOutput:0.298978	Time: 0.330316	InterpSpeed: 1.0
FInterpToOutput:0.310662	Time: 0.346983	InterpSpeed: 1.0
FInterpToOutput:0.322151	Time: 0.36365	InterpSpeed: 1.0
FInterpToOutput:0.333449	Time: 0.380317	InterpSpeed: 1.0
FInterpToOutput:0.344558	Time: 0.396985	InterpSpeed: 1.0
FInterpToOutput:0.355482	Time: 0.413651	InterpSpeed: 1.0
FInterpToOutput:0.366224	Time: 0.430318	InterpSpeed: 1.0
FInterpToOutput:0.376787	Time: 0.446985	InterpSpeed: 1.0
FInterpToOutput:0.387174	Time: 0.463652	InterpSpeed: 1.0

FInterpToConst_Output:0.033334	Time: 0.016667	InterpSpeed: 1.0
FInterpToConst_Output:0.313386	Time: 0.165026	InterpSpeed: 1.0
FInterpToConst_Output:0.218568	Time: 0.191797	InterpSpeed: 1.0
FInterpToConst_Output:0.234363	Time: 0.21308	InterpSpeed: 1.0
FInterpToConst_Output:0.246414	Time: 0.229747	InterpSpeed: 1.0
FInterpToConst_Output:0.263081	Time: 0.246414	InterpSpeed: 1.0
FInterpToConst_Output:0.279748	Time: 0.263081	InterpSpeed: 1.0
FInterpToConst_Output:0.296415	Time: 0.279748	InterpSpeed: 1.0
FInterpToConst_Output:0.313082	Time: 0.296415	InterpSpeed: 1.0
FInterpToConst_Output:0.329749	Time: 0.313082	InterpSpeed: 1.0
FInterpToConst_Output:0.346416	Time: 0.329749	InterpSpeed: 1.0
FInterpToConst_Output:0.363083	Time: 0.346416	InterpSpeed: 1.0
FInterpToConst_Output:0.37975	Time: 0.363083	InterpSpeed: 1.0
FInterpToConst_Output:0.396416	Time: 0.37975	InterpSpeed: 1.0
FInterpToConst_Output:0.413084	Time: 0.396417	InterpSpeed: 1.0
FInterpToConst_Output:0.42975	Time: 0.413083	InterpSpeed: 1.0
FInterpToConst_Output:0.446418	Time: 0.42975	InterpSpeed: 1.0
FInterpToConst_Output:0.463085	Time: 0.446418	InterpSpeed: 1.0
FInterpToConst_Output:0.479751	Time: 0.463084	InterpSpeed: 1.0
FInterpToConst_Output:0.496418	Time: 0.479751	InterpSpeed: 1.0
FInterpToConst_Output:0.513085	Time: 0.496418	InterpSpeed: 1.0

USEFUL UNREAL NODES: INTERPTOCONST

- I wanted to compare the performance more closely, so instead of feeding in deltaTime, I set the deltaTime input to a fixed value of 0.1
- I then ran the test twice, once with a target value of 1, and the other with a target value of 2.

```
FInterpToConst_Output:0.2 Time: 0.1 InterpSpeed: 1.0
FInterpToConst_Output:0.3 Time: 0.2 InterpSpeed: 1.0
FInterpToConst_Output:0.4 Time: 0.3 InterpSpeed: 1.0
FInterpToConst_Output:0.5 Time: 0.4 InterpSpeed: 1.0
FInterpToConst_Output:0.6 Time: 0.5 InterpSpeed: 1.0
FInterpToConst_Output:0.7 Time: 0.6 InterpSpeed: 1.0
FInterpToConst_Output:0.8 Time: 0.7 InterpSpeed: 1.0
FInterpToConst_Output:0.9 Time: 0.8 InterpSpeed: 1.0
FInterpToConst_Output:1.0 Time: 0.9 InterpSpeed: 1.0
FInterpToConst_Output:1.0 Time: 1.0 InterpSpeed: 1.0
FInterpToConst_Output:1.0 Time: 1.1 InterpSpeed: 1.0
```

```
FInterpToConst_Output:0.2 Time: 0.1 InterpSpeed: 1.0
FInterpToConst_Output:0.3 Time: 0.2 InterpSpeed: 1.0
FInterpToConst_Output:0.4 Time: 0.3 InterpSpeed: 1.0
FInterpToConst_Output:0.5 Time: 0.4 InterpSpeed: 1.0
FInterpToConst_Output:0.6 Time: 0.5 InterpSpeed: 1.0
FInterpToConst_Output:0.7 Time: 0.6 InterpSpeed: 1.0
FInterpToConst_Output:0.8 Time: 0.7 InterpSpeed: 1.0
FInterpToConst_Output:0.9 Time: 0.8 InterpSpeed: 1.0
FInterpToConst_Output:1.0 Time: 0.9 InterpSpeed: 1.0
FInterpToConst_Output:1.1 Time: 1.0 InterpSpeed: 1.0
FInterpToConst_Output:1.2 Time: 1.1 InterpSpeed: 1.0
FInterpToConst_Output:1.3 Time: 1.2 InterpSpeed: 1.0
FInterpToConst_Output:1.4 Time: 1.3 InterpSpeed: 1.0
FInterpToConst_Output:1.5 Time: 1.4 InterpSpeed: 1.0
FInterpToConst_Output:1.6 Time: 1.5 InterpSpeed: 1.0
FInterpToConst_Output:1.7 Time: 1.6 InterpSpeed: 1.0
FInterpToConst_Output:1.8 Time: 1.7 InterpSpeed: 1.0
FInterpToConst_Output:1.9 Time: 1.8 InterpSpeed: 1.0
FInterpToConst_Output:2.0 Time: 1.9 InterpSpeed: 1.0
FInterpToConst_Output:2.0 Time: 2.0 InterpSpeed: 1.0
FInterpToConst_Output:2.0 Time: 2.1 InterpSpeed: 1.0
FInterpToConst_Output:2.0 Time: 2.2 InterpSpeed: 1.0
```

USEFUL UNREAL NODES: INTERPTOCONST

- This last test was interesting because it showed two distinct pieces of information:
- 1. The rate was constant, regardless of the target. This is different from how a Lerp normally works as the target value will change the rate of change.
- 2. The rate appears to change each tick at a rate of $\text{deltaTime} * \text{InterpSpeed}$

```
FInterpToConst_Output:0.2 Time: 0.1 InterpSpeed: 1.0
FInterpToConst_Output:0.3 Time: 0.2 InterpSpeed: 1.0
FInterpToConst_Output:0.4 Time: 0.3 InterpSpeed: 1.0
FInterpToConst_Output:0.5 Time: 0.4 InterpSpeed: 1.0
FInterpToConst_Output:0.6 Time: 0.5 InterpSpeed: 1.0
FInterpToConst_Output:0.7 Time: 0.6 InterpSpeed: 1.0
FInterpToConst_Output:0.8 Time: 0.7 InterpSpeed: 1.0
FInterpToConst_Output:0.9 Time: 0.8 InterpSpeed: 1.0
FInterpToConst_Output:1.0 Time: 0.9 InterpSpeed: 1.0
FInterpToConst_Output:1.0 Time: 1.0 InterpSpeed: 1.0
FInterpToConst_Output:1.0 Time: 1.1 InterpSpeed: 1.0
```

```
FInterpToConst_Output:0.2 Time: 0.1 InterpSpeed: 1.0
FInterpToConst_Output:0.3 Time: 0.2 InterpSpeed: 1.0
FInterpToConst_Output:0.4 Time: 0.3 InterpSpeed: 1.0
FInterpToConst_Output:0.5 Time: 0.4 InterpSpeed: 1.0
FInterpToConst_Output:0.6 Time: 0.5 InterpSpeed: 1.0
FInterpToConst_Output:0.7 Time: 0.6 InterpSpeed: 1.0
FInterpToConst_Output:0.8 Time: 0.7 InterpSpeed: 1.0
FInterpToConst_Output:0.9 Time: 0.8 InterpSpeed: 1.0
FInterpToConst_Output:1.0 Time: 0.9 InterpSpeed: 1.0
FInterpToConst_Output:1.1 Time: 1.0 InterpSpeed: 1.0
FInterpToConst_Output:1.2 Time: 1.1 InterpSpeed: 1.0
FInterpToConst_Output:1.3 Time: 1.2 InterpSpeed: 1.0
FInterpToConst_Output:1.4 Time: 1.3 InterpSpeed: 1.0
FInterpToConst_Output:1.5 Time: 1.4 InterpSpeed: 1.0
FInterpToConst_Output:1.6 Time: 1.5 InterpSpeed: 1.0
FInterpToConst_Output:1.7 Time: 1.6 InterpSpeed: 1.0
FInterpToConst_Output:1.8 Time: 1.7 InterpSpeed: 1.0
FInterpToConst_Output:1.9 Time: 1.8 InterpSpeed: 1.0
FInterpToConst_Output:2.0 Time: 1.9 InterpSpeed: 1.0
FInterpToConst_Output:2.0 Time: 2.0 InterpSpeed: 1.0
FInterpToConst_Output:2.0 Time: 2.1 InterpSpeed: 1.0
FInterpToConst_Output:2.0 Time: 2.2 InterpSpeed: 1.0
```

USEFUL UNREAL NODES: INTERPTOCONST

- To confirm this theory, I ran one more test.
- For the output on the left, I set the deltaTime to 0.2 and the speed to 1.
- For the output on the right, I set the deltaTime to 0.1 and the speed to 2.
- As you can see, the output is the same, confirming my theory.

```
FInterpToConst_Output:0.4 Time: 0.1 InterpSpeed: 1.0  
FInterpToConst_Output:0.6 Time: 0.2 InterpSpeed: 1.0  
FInterpToConst_Output:0.8 Time: 0.3 InterpSpeed: 1.0  
FInterpToConst_Output:1.0 Time: 0.4 InterpSpeed: 1.0  
FInterpToConst_Output:1.0 Time: 0.5 InterpSpeed: 1.0  
FInterpToConst_Output:1.0 Time: 0.6 InterpSpeed: 1.0
```

```
FInterpToConst_Output:0.4 Time: 0.1 InterpSpeed: 2.0  
FInterpToConst_Output:0.6 Time: 0.2 InterpSpeed: 2.0  
FInterpToConst_Output:0.8 Time: 0.3 InterpSpeed: 2.0  
FInterpToConst_Output:1.0 Time: 0.4 InterpSpeed: 2.0  
FInterpToConst_Output:1.0 Time: 0.5 InterpSpeed: 2.0  
FInterpToConst_Output:1.0 Time: 0.6 InterpSpeed: 2.0  
FInterpToConst_Output:1.0 Time: 0.7 InterpSpeed: 2.0  
FInterpToConst_Output:1.0 Time: 0.8 InterpSpeed: 2.0
```

BACK TO INTERPTO:

- This made me interested in how the InterpTo Node would function with a fixed deltaTime


```
FInterpToOutput:0.36 Time: 0.1 InterpSpeed: 2.0
FInterpToOutput:0.488 Time: 0.2 InterpSpeed: 2.0
FInterpToOutput:0.5904 Time: 0.3 InterpSpeed: 2.0
FInterpToOutput:0.67232 Time: 0.4 InterpSpeed: 2.0
FInterpToOutput:0.737856 Time: 0.5 InterpSpeed: 2.0
FInterpToOutput:0.790285 Time: 0.6 InterpSpeed: 2.0
FInterpToOutput:0.832228 Time: 0.7 InterpSpeed: 2.0
FInterpToOutput:0.865782 Time: 0.8 InterpSpeed: 2.0
FInterpToOutput:0.892626 Time: 0.9 InterpSpeed: 2.0
FInterpToOutput:0.914101 Time: 1.0 InterpSpeed: 2.0
FInterpToOutput:0.93128 Time: 1.1 InterpSpeed: 2.0
FInterpToOutput:0.945024 Time: 1.2 InterpSpeed: 2.0
FInterpToOutput:0.95602 Time: 1.3 InterpSpeed: 2.0
FInterpToOutput:0.964816 Time: 1.4 InterpSpeed: 2.0
FInterpToOutput:0.971852 Time: 1.5 InterpSpeed: 2.0
FInterpToOutput:0.977482 Time: 1.6 InterpSpeed: 2.0
FInterpToOutput:0.981986 Time: 1.7 InterpSpeed: 2.0
FInterpToOutput:0.985588 Time: 1.8 InterpSpeed: 2.0
FInterpToOutput:0.988471 Time: 1.9 InterpSpeed: 2.0
FInterpToOutput:0.990777 Time: 2.0 InterpSpeed: 2.0
FInterpToOutput:0.992621 Time: 2.1 InterpSpeed: 2.0
FInterpToOutput:0.994097 Time: 2.2 InterpSpeed: 2.0
FInterpToOutput:0.995278 Time: 2.3 InterpSpeed: 2.0
FInterpToOutput:0.996222 Time: 2.4 InterpSpeed: 2.0
FInterpToOutput:0.996978 Time: 2.5 InterpSpeed: 2.0
FInterpToOutput:0.997582 Time: 2.6 InterpSpeed: 2.0
FInterpToOutput:0.998066 Time: 2.7 InterpSpeed: 2.0
FInterpToOutput:0.998453 Time: 2.799999 InterpSpeed: 2.0
FInterpToOutput:0.998762 Time: 2.899999 InterpSpeed: 2.0
FInterpToOutput:0.99901 Time: 2.999999 InterpSpeed: 2.0
FInterpToOutput:0.999208 Time: 3.099999 InterpSpeed: 2.0
FInterpToOutput:0.999366 Time: 3.199999 InterpSpeed: 2.0
FInterpToOutput:0.999493 Time: 3.299999 InterpSpeed: 2.0
FInterpToOutput:0.999594 Time: 3.399999 InterpSpeed: 2.0
FInterpToOutput:0.999675 Time: 3.499999 InterpSpeed: 2.0
FInterpToOutput:0.99974 Time: 3.599999 InterpSpeed: 2.0
FInterpToOutput:0.999792 Time: 3.699999 InterpSpeed: 2.0
FInterpToOutput:0.999834 Time: 3.799999 InterpSpeed: 2.0
FInterpToOutput:0.999867 Time: 3.899998 InterpSpeed: 2.0
FInterpToOutput:0.999894 Time: 3.999998 InterpSpeed: 2.0
FInterpToOutput:0.999915 Time: 4.099998 InterpSpeed: 2.0
FInterpToOutput:1.0 Time: 4.199998 InterpSpeed: 2.0
FInterpToOutput:1.0 Time: 4.299998 InterpSpeed: 2.0
FInterpToOutput:1.0 Time: 4.399998 InterpSpeed: 2.0
```

```
FInterpToOutput:0.64 Time: 0.1 InterpSpeed: 4.0
FInterpToOutput:0.784 Time: 0.2 InterpSpeed: 4.0
FInterpToOutput:0.8704 Time: 0.3 InterpSpeed: 4.0
FInterpToOutput:0.92224 Time: 0.4 InterpSpeed: 4.0
FInterpToOutput:0.953344 Time: 0.5 InterpSpeed: 4.0
FInterpToOutput:0.972006 Time: 0.6 InterpSpeed: 4.0
FInterpToOutput:0.983204 Time: 0.7 InterpSpeed: 4.0
FInterpToOutput:0.989922 Time: 0.8 InterpSpeed: 4.0
FInterpToOutput:0.993953 Time: 0.9 InterpSpeed: 4.0
FInterpToOutput:0.996372 Time: 1.0 InterpSpeed: 4.0
FInterpToOutput:0.997823 Time: 1.1 InterpSpeed: 4.0
FInterpToOutput:0.998694 Time: 1.2 InterpSpeed: 4.0
FInterpToOutput:0.999216 Time: 1.3 InterpSpeed: 4.0
FInterpToOutput:0.99953 Time: 1.4 InterpSpeed: 4.0
FInterpToOutput:0.999718 Time: 1.5 InterpSpeed: 4.0
FInterpToOutput:0.999831 Time: 1.6 InterpSpeed: 4.0
FInterpToOutput:0.999898 Time: 1.7 InterpSpeed: 4.0
FInterpToOutput:0.999939 Time: 1.8 InterpSpeed: 4.0
FInterpToOutput:1.0 Time: 1.9 InterpSpeed: 4.0
FInterpToOutput:1.0 Time: 2.0 InterpSpeed: 4.0
FInterpToOutput:1.0 Time: 2.1 InterpSpeed: 4.0
FInterpToOutput:1.0 Time: 2.2 InterpSpeed: 4.0
FInterpToOutput:1.0 Time: 2.3 InterpSpeed: 4.0
FInterpToOutput:1.0 Time: 2.4 InterpSpeed: 4.0
FInterpToOutput:1.0 Time: 2.5 InterpSpeed: 4.0
FInterpToOutput:1.0 Time: 2.6 InterpSpeed: 4.0
FInterpToOutput:1.0 Time: 2.7 InterpSpeed: 4.0
```

DIFFERENT SPEEDS, SAME TARGET

```
FInterpToOutput:0.36 Time: 0.1 InterpSpeed: 2.0
FInterpToOutput:0.488 Time: 0.2 InterpSpeed: 2.0
FInterpToOutput:0.5904 Time: 0.3 InterpSpeed: 2.0
FInterpToOutput:0.67232 Time: 0.4 InterpSpeed: 2.0
FInterpToOutput:0.737856 Time: 0.5 InterpSpeed: 2.0
FInterpToOutput:0.790285 Time: 0.6 InterpSpeed: 2.0
FInterpToOutput:0.832228 Time: 0.7 InterpSpeed: 2.0
FInterpToOutput:0.865782 Time: 0.8 InterpSpeed: 2.0
FInterpToOutput:0.892626 Time: 0.9 InterpSpeed: 2.0
FInterpToOutput:0.914101 Time: 1.0 InterpSpeed: 2.0
FInterpToOutput:0.93128 Time: 1.1 InterpSpeed: 2.0
FInterpToOutput:0.945024 Time: 1.2 InterpSpeed: 2.0
FInterpToOutput:0.95602 Time: 1.3 InterpSpeed: 2.0
FInterpToOutput:0.964816 Time: 1.4 InterpSpeed: 2.0
FInterpToOutput:0.971852 Time: 1.5 InterpSpeed: 2.0
FInterpToOutput:0.977482 Time: 1.6 InterpSpeed: 2.0
FInterpToOutput:0.981986 Time: 1.7 InterpSpeed: 2.0
FInterpToOutput:0.985588 Time: 1.8 InterpSpeed: 2.0
FInterpToOutput:0.988471 Time: 1.9 InterpSpeed: 2.0
FInterpToOutput:0.990777 Time: 2.0 InterpSpeed: 2.0
FInterpToOutput:0.992621 Time: 2.1 InterpSpeed: 2.0
FInterpToOutput:0.994097 Time: 2.2 InterpSpeed: 2.0
FInterpToOutput:0.995278 Time: 2.3 InterpSpeed: 2.0
FInterpToOutput:0.996222 Time: 2.4 InterpSpeed: 2.0
FInterpToOutput:0.996978 Time: 2.5 InterpSpeed: 2.0
FInterpToOutput:0.997582 Time: 2.6 InterpSpeed: 2.0
FInterpToOutput:0.998066 Time: 2.7 InterpSpeed: 2.0
FInterpToOutput:0.998453 Time: 2.799999 InterpSpeed: 2.0
FInterpToOutput:0.998762 Time: 2.899999 InterpSpeed: 2.0
FInterpToOutput:0.99901 Time: 2.999999 InterpSpeed: 2.0
FInterpToOutput:0.999208 Time: 3.099999 InterpSpeed: 2.0
FInterpToOutput:0.999366 Time: 3.199999 InterpSpeed: 2.0
FInterpToOutput:0.999493 Time: 3.299999 InterpSpeed: 2.0
FInterpToOutput:0.999594 Time: 3.399999 InterpSpeed: 2.0
FInterpToOutput:0.999675 Time: 3.499999 InterpSpeed: 2.0
FInterpToOutput:0.99974 Time: 3.599999 InterpSpeed: 2.0
FInterpToOutput:0.999792 Time: 3.699999 InterpSpeed: 2.0
FInterpToOutput:0.999834 Time: 3.799999 InterpSpeed: 2.0
FInterpToOutput:0.999867 Time: 3.899998 InterpSpeed: 2.0
FInterpToOutput:0.999894 Time: 3.999998 InterpSpeed: 2.0
FInterpToOutput:0.999915 Time: 4.099998 InterpSpeed: 2.0
FInterpToOutput:1.0 Time: 4.199998 InterpSpeed: 2.0
FInterpToOutput:1.0 Time: 4.299998 InterpSpeed: 2.0
FInterpToOutput:1.0 Time: 4.399998 InterpSpeed: 2.0
```

```
FInterpToOutput:0.72 Time: 0.1 InterpSpeed: 2.0
FInterpToOutput:0.976 Time: 0.2 InterpSpeed: 2.0
FInterpToOutput:1.1808 Time: 0.3 InterpSpeed: 2.0
FInterpToOutput:1.34464 Time: 0.4 InterpSpeed: 2.0
FInterpToOutput:1.475712 Time: 0.5 InterpSpeed: 2.0
FInterpToOutput:1.58057 Time: 0.6 InterpSpeed: 2.0
FInterpToOutput:1.664456 Time: 0.7 InterpSpeed: 2.0
FInterpToOutput:1.731565 Time: 0.8 InterpSpeed: 2.0
FInterpToOutput:1.785252 Time: 0.9 InterpSpeed: 2.0
FInterpToOutput:1.828201 Time: 1.0 InterpSpeed: 2.0
FInterpToOutput:1.862561 Time: 1.1 InterpSpeed: 2.0
FInterpToOutput:1.890049 Time: 1.2 InterpSpeed: 2.0
FInterpToOutput:1.912039 Time: 1.3 InterpSpeed: 2.0
FInterpToOutput:1.929631 Time: 1.4 InterpSpeed: 2.0
FInterpToOutput:1.943705 Time: 1.5 InterpSpeed: 2.0
FInterpToOutput:1.954964 Time: 1.6 InterpSpeed: 2.0
FInterpToOutput:1.963971 Time: 1.7 InterpSpeed: 2.0
FInterpToOutput:1.971177 Time: 1.8 InterpSpeed: 2.0
FInterpToOutput:1.976941 Time: 1.9 InterpSpeed: 2.0
FInterpToOutput:1.981553 Time: 2.0 InterpSpeed: 2.0
FInterpToOutput:1.985243 Time: 2.1 InterpSpeed: 2.0
FInterpToOutput:1.988194 Time: 2.2 InterpSpeed: 2.0
FInterpToOutput:1.990555 Time: 2.3 InterpSpeed: 2.0
FInterpToOutput:1.992444 Time: 2.4 InterpSpeed: 2.0
FInterpToOutput:1.993955 Time: 2.5 InterpSpeed: 2.0
FInterpToOutput:1.995164 Time: 2.6 InterpSpeed: 2.0
FInterpToOutput:1.996131 Time: 2.7 InterpSpeed: 2.0
FInterpToOutput:1.996905 Time: 2.799999 InterpSpeed: 2.0
FInterpToOutput:1.997524 Time: 2.899999 InterpSpeed: 2.0
FInterpToOutput:1.998019 Time: 2.999999 InterpSpeed: 2.0
FInterpToOutput:1.998415 Time: 3.099999 InterpSpeed: 2.0
FInterpToOutput:1.998732 Time: 3.199999 InterpSpeed: 2.0
FInterpToOutput:1.998986 Time: 3.299999 InterpSpeed: 2.0
FInterpToOutput:1.999189 Time: 3.399999 InterpSpeed: 2.0
FInterpToOutput:1.999351 Time: 3.499999 InterpSpeed: 2.0
FInterpToOutput:1.999481 Time: 3.599999 InterpSpeed: 2.0
FInterpToOutput:1.999585 Time: 3.699999 InterpSpeed: 2.0
FInterpToOutput:1.999668 Time: 3.799999 InterpSpeed: 2.0
FInterpToOutput:1.999734 Time: 3.899998 InterpSpeed: 2.0
FInterpToOutput:1.999787 Time: 3.999998 InterpSpeed: 2.0
FInterpToOutput:1.99983 Time: 4.099998 InterpSpeed: 2.0
FInterpToOutput:1.999864 Time: 4.199998 InterpSpeed: 2.0
FInterpToOutput:1.999891 Time: 4.299998 InterpSpeed: 2.0
FInterpToOutput:1.999913 Time: 4.399998 InterpSpeed: 2.0
FInterpToOutput:2.0 Time: 4.499998 InterpSpeed: 2.0
FInterpToOutput:2.0 Time: 4.599998 InterpSpeed: 2.0
FInterpToOutput:2.0 Time: 4.699998 InterpSpeed: 2.0
```

DIFFERENT
TARGETS,
SAME SPEED

INTERPTO QUESTIONS:

- So looking closely at the previous 2 slides, what do you notice about the way InterpTo works?
- What is different and what is similar between the two nodes?
- Does changing the speed effect how long it takes to reach its target value?
- Does changing the target value effect how long it takes to reach its target value?

INTERPTO SUMMARY:

- As you can see, changing the speed of the InterpTo node doesn't effect how long it takes to reach the target, however it increases the amount the current value changes each frame.
- That is, InterpTo will reach any target in the same time if it has the same speed.
- InterpToConstant will have different behaviour and will move at a fixed rate towards a target. This means changing the target will change the amount of time it takes to reach that target value.
- As you can see, these nodes can be a little tricky to work with, especially since they operate in fundamentally different ways from each other, despite having a similar looking interface and variables.

CLOSING NOTES:

- I hope this exercise was valuable not only for its exploration of these nodes but also for its demonstration of the scientific process in a practical sense.
- What we essentially did here was form a working hypothesis about how we ASSUMED the nodes SHOULD work, then tested those theories on data.
- We also controlled for outside variables (such as the deltaTime and the weird glitches it caused when I first ran the scene) so that we could get more accurate data to update our model of the nodes.
- You will often have to do this sort of thing when working in games. Rarely will the documentation give you the full amount of information you need and you should always test your assumptions before getting too invested in a process.